



Vehicle Data Monetization Streams – Disrupting traditional Automotive Value Chain

Abstract:

The automotive industry is seeing a dynamic shift in its revenue channels. Data is steadily becoming an invaluable commodity and the connected car is enabling unique revenue propositions. Connectivity and shared mobility ecosystems are facilitating this expansion in use cases and services. OEMs in conjunction with data aggregators are making this data more widely available, allowing other industries such as insurance companies and fleet management companies offer advanced value driven services.

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Introduction

Automotive sector is experiencing disruption from three major global megatrends, leading to definite shift in markets and revenue streams, diffusion of technology and entry of new players in the ecosystem. (Electrification is also one of the major trends, which is not discussed in this paper)

- Connected Cars – Internet and telematics-based connectivity capabilities inside vehicle
- Autonomous Vehicles – Autonomous driving capabilities that include technologies such as Autonomous Driving Assistance Systems (ADAS), LiDAR etc
- Shared Mobility Ecosystem – Transportation resources such as passenger vehicles will be shared amongst users. For example, Mercedes Benz offers a pay-as-you-drive model for users to rent its vehicles primarily in Europe and US markets.

These megatrends are leading to an unprecedented explosion in car-generated digital data, with significant implications in automotive and allied industries such as Insurance, Telecom, Semiconductor etc. For example, in 2020, 64 million cars were enabled with internet and connected capabilities, with each car generating around 200 to 300 mb data from their telematics unit alone. By 2025, the number of connected cars will grow to over 250 million with each telematics unit generating over 500 mb data. With the proliferation of data, monetizing this data collected from cars has become a key theme mainly for the automotive OEMs and Telematics & Mobility players.



Types of Car Data and Monetization Opportunities

Car data consists of different types of data points from various sources. For example, in V2X communications, vehicles interact with smart city networks and generate a location data trail. Traditionally, connectivity was included in vehicles to increase road safety and in-vehicle automation. However, with data proliferation, many industry bodies see value in harnessing this data for other use-cases. For example, National Road Authority uses this vehicle data for traffic management – optimizing traffic, reducing congestion and city planning. The connected car enables multiple use cases, which can range from bringing in recurring revenues to reducing overall costs in the ecosystem.

The following table gives an insight into some of the most common types of vehicle data, its commercial use cases (now and in future) and the players that can utilize various data points to generate new revenue streams.

DATA TYPES	DATA MONETIZATION ATTRIBUTES	TODAY	BY 2025	COMPANIES THAT WILL BUY/UTILIZE VEHICLE DATA
PERSONAL DATA Driver behavior, Entertainment (android car play), social media, identity	 Personal data, preferences (e.g., driver/passengers' identity, preferred radio station)	 Vehicle memory based on key presence at entry	- E-commerce in the car - Targeted advertisements	Retail & E-commerce : Costco, Amazon, Walmart, 7-11 Social Media : Spotify, Facebook
AMBIENT (V2X) DATA Safety, weather conditions, V2V, V2I, mobility, traffic	 External road and environmental conditions (e.g., ice warning on the road from ESP, fog from camera/sensors' feed)	 Real-time maps	Preventive safety car Adaptation, Live road conditions reports	Government Agencies & TSP players: NRA, Tomtom, Geotab, Traffic Management MaaS Players: Hitachi, Siemens, Alstom
VEHICLE DATA Usage data, quality control, Vehicle Diagnostics, Vehicle Maintenance, Emergency Support	 Technical status of the vehicle (e.g., oil temperature, airbag Deployment)	 - Car repair diagnostics - Automatic emergency call (e-call)	Predictive, remote service booking, remote control, parking assist	Tier 1 & Aftermarket Service Players : 3M, Faurecia, Denso etc Insurance Providers : Allstate, Aviva, BNP
	 Vehicle usage (e.g., speed, location, battery status)	 - PAYD insurance - Toll/road tax payment	- Home Automation - Garage Door - EV Charging	Smart Home : ABB, Honeywell, Google

Table 1: Examples on Vehicle Data Types, Use Cases and Organizations utilizing vehicle data

Vehicle data has a potential to generate exponential value, as it is possible to re-sell the same data to various industry bodies to solve a horde of use cases.

Mckinsey estimates a revenue of USD 450-700 billion by 2030 from the top thirty use cases of vehicle data such as insurance, fleet management, parking, to name a few.



OEM's critical role in Automotive Data Ecosystem

Until recently (2013-14), OEMs did not have a business case to share car data as vehicles were not connected since integrating connectivity was expensive and users were not ready to pay for connected data use cases. The collection of data mostly relied on aftermarket telematics devices (Tomtom, Geotab etc). Today, with advancements in semiconductor and connectivity technologies, the hardware and integration cost of connectivity has reduced significantly, leading to the commoditization of sensors. Cars are now being connected at a larger scale and are equipped with various sensors. The onset of connectivity and shared mobility ecosystem has enabled many stakeholders who wish to collect reliable car data at a moderate cost. Many players, even from non-automotive industry, are interested in using the data to provide more advanced services and products to the users. As a result, multiple players need to in turn send data to multiple other players ($n \times m$), creating a very complex data sharing environment. With a complex ecosystem comes multiple challenges as well:

- OEMs were positioned to gain control over all the data moving in and out of the whole automotive value chain
- This created avenues for information asymmetry and lack of control over data for other players (example – after services), leading to monopoly of data by OEMs
- Also, most data associated with connected vehicles fall into personal data category (e.g. geolocation data; biometric data; etc.) and lead to risks of unlawful processing of personal data by violating GDPR norms.



Emergence of data aggregator because of Regulatory requirements

In 1980s, vehicle monitoring systems were developed in response to regulate emissions from vehicles. California Air Resources Board (CARB) mandated OBD port (1991) as the electronic access to vehicle data. This was further incorporated as a part of UN Regulations, leading to development of the Standard OBD2 connector (1996). OBD2 connector was mandated for all vehicles as way of independent, direct, real-time and free of charge access to vehicle data. We can see a very similar story where regulations over data sharing by European Union led to emergence of data aggregators.

To develop a level playing field and to facilitate fair access to data for all players, EU developed a new regulation-based framework for car data access. In accordance with the European Union's Regulation (EU) 2018/858, OEMs must be ready to share connected car data with third parties (data aggregators) in a standardized format, which can be accessed by anybody who wishes to use this data. This resulted in global ISO 20078 guidelines of data sharing.

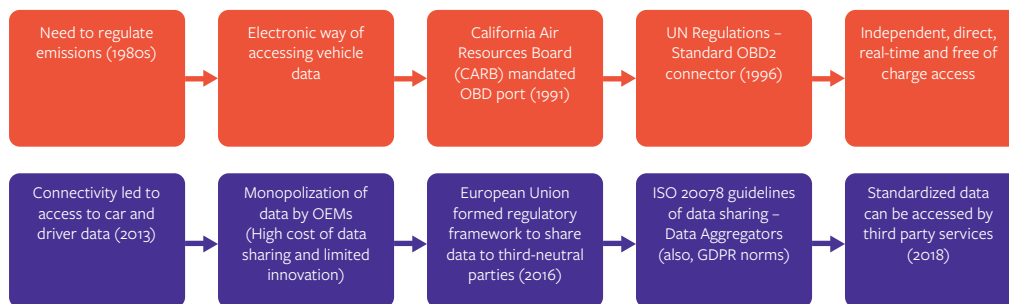


Table 2: OBD Evolution Vs Data Aggregator Evolution

The extended vehicle concept diagram is shown below:

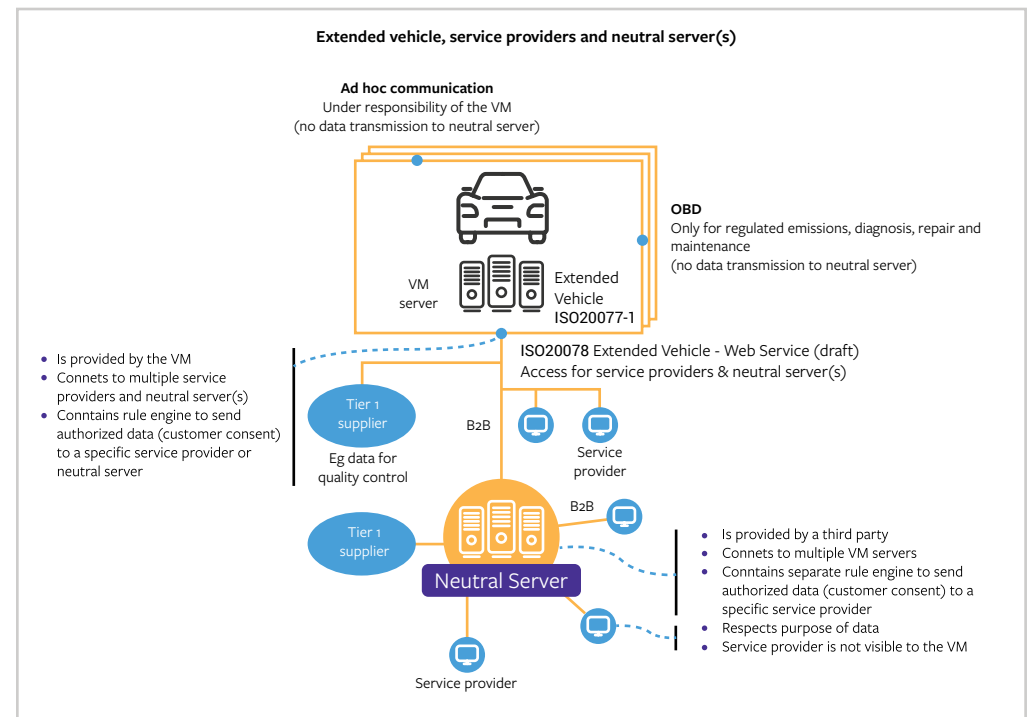


Figure 1 : Extended Vehicle Concept from EU car data access regulation (Ref : ACEA Position Paper)



Data Aggregator Business Model

The main role of data aggregators is to act as a data market place for third party players (insurance, fleet management, government agencies), where raw vehicle data from multiple OEMs is processed, standardized and shared across various entities using different access APIs. Examples of such data aggregators include startups like Wejo, High Mobility, Caruso, Otonomo.

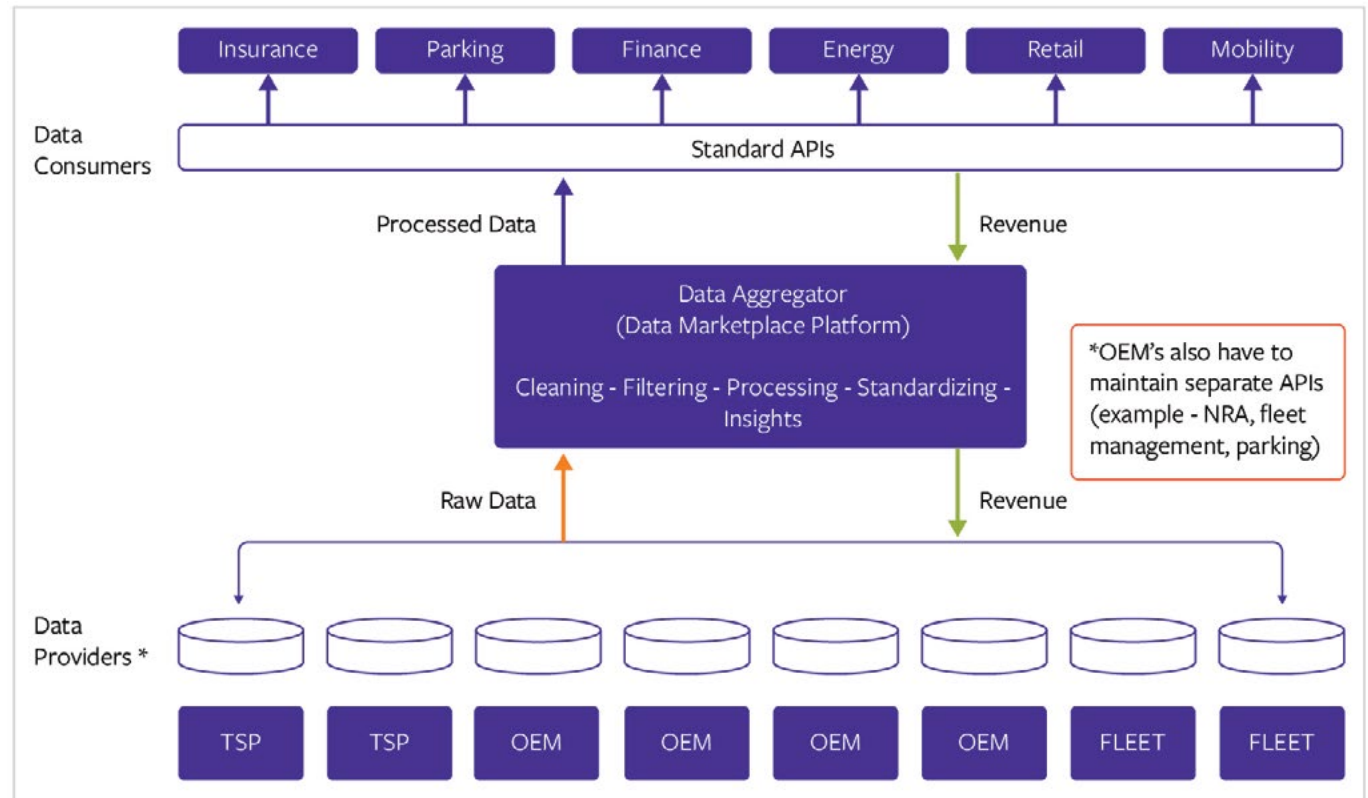


Figure 2 : Data Aggregator Business Model



As per PTOLEMUS consulting group report, the data aggregator market is further segmented into four markets:

- **Data Analysis:** Data processing (cleaning, filtering, insights) and analytics (predictive and prescriptive) services
- **Data Exchange:** Platform to access multibrand in-vehicle data, data exchange management, bill and revenue management
- **Service and Product Development:** Application environment, SDK for developer community
- **Services and Product Offerings:** B2B, B2C and B2B2C customers

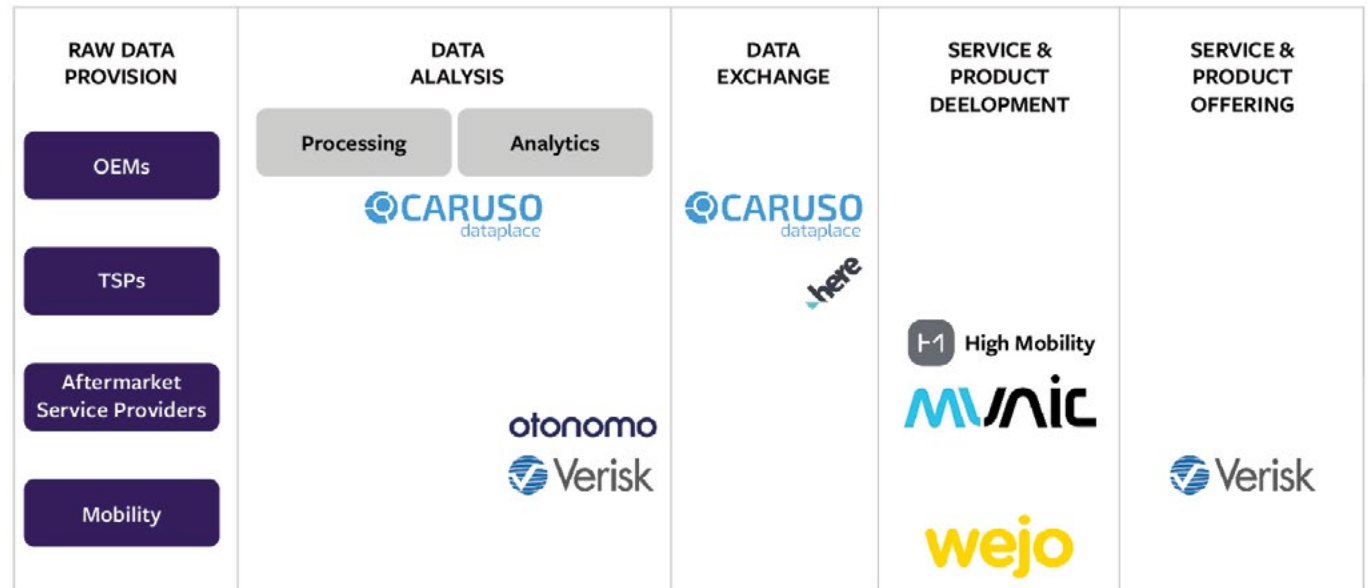


Table 3: Data Aggregator Market Segmentation (Ref: [Ptolemus.com](https://ptolemus.com))

Each data aggregator has uniquely positioned itself across different market segments:

Data aggregators package the processed data collected from OEMs and sell it to different third party service providers. For example, an insurance provider decides the vehicle insurance premium to users based on vehicle's accident history and driving behaviour. The insurance provider accesses the vehicle history through

the data aggregator's marketplace platform. The price of data is generally governed by revenue sharing agreement between data aggregator and OEMs. Few examples of data aggregators and their pricing models are given below.



Data Aggregator	Affiliation with OEMs	Pricing Model (per vehicle/month)	Top Use Cases (data sharing with third parties)
Caruso	Audi, BMW, Ford, Mercedes-Benz, Mini, VW	\$ 0.35 to \$ 10	Vehicle Health and Maintenance, Vehicle Movement and Location
High Mobility	Audi, BMW, Ford, Mercedes-Benz, Mini, VW	\$ 8 for BMW \$2.5 for Benz	EV Charging, Insurance, Vehicle Maintenance
Wejo	Daimler, General Motors, Ford	Not disclosed	Traffic, Insurance, Fleet, Parking
Otonomo	BMW, Mercedes-Benz	\$60 for 1M points or \$60 for 20K trips	Smart Cities, Traffic Management, Safety & Emergency

Table 4: Data Aggregator and their Pricing Models

Data aggregation ecosystem helps OEMs and Telematics & Mobility players widen their access to multiple entities, helping the players gain benefits of network effects of mobility ecosystem and, hence additional and recurring revenue streams.

A user journey is mapped below to understand how an OEM generates additional revenue from a parking application.

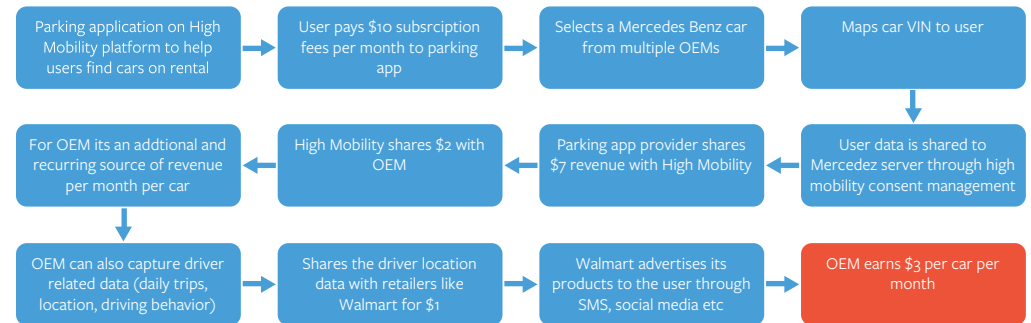


Figure 3 : User journey in mobility ecosystem and benefits for OEMs

We can estimate the market size of these data aggregators based on user journey and total number of cars registered their platform. As per calculations given in the table below, this market (data aggregators, OEM, Telematics and Mobility players) is poised to grow at CAGR of 200% which will reach ~ \$ 38 billion by 2026.

Year	Accessible Cars (M)	Active Cars (M)	Percentage	Uses per month (USD)	Total Revenue (M)
2020	69	10.3	0.15	8.00	988.80
2021	90	27	0.30	8.96	2903.04
2022	136	54.4	0.40	9.86	6434.00
2023	166	91.3	0.55	10.84	11878.06
2024	195	126.75	0.65	11.93	18139.08
2025	200	160	0.80	13.12	25187.21
2026	224	212.8	0.95	14.43	36848.88

*Revenue generated from data monetization to third parties and for connected cars only

Table 5 : Revenue estimates for data aggregator ecosystem



Key Challenges from Data Aggregator ecosystem for automotive OEMs

Although OEMs are poised to gain significantly from data monetization, the data aggregator ecosystem brings various challenges to OEMs. The challenges are discussed below in detail:

- **Mass Customization:** The data generated from vehicles is based on different car models, versions, geographies and local regulations leading to a very high variety and volume of data. As current data aggregator ecosystems have limited bandwidth, cost of customizations will be very high for OEMs
- **Continuously changing technologies and protocols:** Data aggregators are facing difficulties in coping with continuously changing communication technologies and protocols (Example: the onset of 5G)
- **Multiple data sharing avenues:** OEMs have to maintain multiple data sharing avenues such as:
 - a. Various data aggregators for different use cases
 - b. Mandatory data sharing with Government agencies
 - c. Internal business units (insurance and fleet), leading to high operational costs of data sharing and management.

- **Cost of development and maintenance of internal teams:** OEMs have to maintain internal teams for data management and standardization, which will be directly shared with third parties for different use cases. Technological maturity levels of OEMs vary in every region. Hence, some OEMs have to develop and maintain internal teams for data management. (for example - BMW plans to expand its cloud and data team to 5000 people in the coming years) This leads to very high OPEX for OEMs.

With above challenges, OEMs find it difficult to economize scale in terms of data management as growth is outweighed by high operational costs and loss of control to data aggregators.



Conclusion

It is apparent that automotive industry's topline are shifting from unit sales to data-based recurring revenues. As the new landscape is evolving (based on mobility and connectivity), the overall automotive value chain is being disrupted by new entrants such as data aggregators and high-tech startups. Car data monetization will eventually shift the revenue pools to new entrants, like data aggregators, based on their advancements in technological capabilities. To match the growing demands of data, capture new revenue streams, and develop low-cost data sharing avenues, OEMs (also telematics and mobility players) should:

- Develop long term data monetization strategies and steer the organization towards a transformational change
- Invest in advanced data management capabilities and infrastructure
- Define data-related use cases and business models in detail and build an ecosystem of partners to deliver on the use cases
- Partner with service players with expertise in Connectivity, Telematics and Data Management capabilities

About the Author

Rohit Anand is currently leading IoT and Analytics Portfolio at Sasken Technologies Limited. An IIM Bangalore alumnus, he has worked extensively on formulation of Growth strategy and consults senior executives on Transformation initiatives across multiple sectors

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





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